# Project Report - Day 9

**Title: Engineering a Robust Input-Output Pipeline for RAG**

## Objective

The objective of this task is to build a robust Retrieval-Augmented Generation (RAG) pipeline that can handle real-world scenarios. The system focuses on validating user input, retrieving relevant information from a document using embeddings, and generating accurate responses while tracking system performance through latency logging. The goal is to make the pipeline reliable, efficient, and production-ready..

## Libraries Used

1. **pathlib**
   - Used to build a reliable file path for loading corpus.txt from the same folder as the script.
2. **sentence-transformers**
   - Used to convert text data (documents and queries) into vector embeddings for semantic search.
3. **sklearn (cosine_similarity)**
   - Used to measure similarity between query embeddings and document embeddings to retrieve the most relevant context.
4. **dotenv**
   - Used to load environment variables from the .env file.
5. **Time**
   - Used to calculate retrieval time and generation time for performance logging.
6. **Google-genai**
   • Used to integrate Gemini API for generating clean and context-based responses.

## Errors

1. **Invalid Input handling issue**
    - **Issue:** The system accepted empty or meaningless inputs like special characters.
    - **Solution:** Input validation was added to reject empty queries and queries with only special characters.
2. **.env loading issue**
    - **Issue:** The environment file was not loaded reliably in all execution cases.
    - **Solution:** The script was changed to avoid depending on .env for the final version.
3. **Long Query (Token Stuffing) Issue**
    - **Issue:** Very long queries caused inefficient processing and potential performance issues.
    - **Solution:** A character limit (300 characters) was applied to restrict excessively long inputs
4. **Incorrect Retrieval (Wrong Context)**
    - **Issue:** The system retrieved irrelevant chunks due to poor chunking and semantic mismatch.
    - **Solution:** Improved chunking strategy and used Top-K retrieval to improve accuracy.
5. **Mixed or Noisy Output**
    - **Issue:** Retrieved chunks contained multiple topics leading to incorrect or noisy responses.
    - **Solution:** Context cleaning and better chunk splitting were implemented to isolate meaningful sections.

## Output

```
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
User Query: Who is the CEO of NexusVortex?
Response:
 Answer not found in context.
User Query: What size must the /echo partition be?
Response:
 Exactly 512MB
User Query: What is Stasis Mode?
Response:
 Stasis Mode is a fail-safe state where the Pulse-Core throttles all non-essential I/O to 5% of maximum capacity to prevent Data Bleed, which occurs when unsynchronized sh
ards write conflicting data to the V-FAT32 file system.
User Query: What must VORTEX_PATH be mapped to?
Response:
 /pulse/bin
User Query: Can I install this OS on a normal i7 laptop?
Response:
 Answer not found in context.
User Query: What command fixes UI ghosting?
Response:
 vortex --realign-pulse
User Query:
```